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        AI LOGICAL REASONING - ASSESSMENT 1
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CASE STUDY 1 - SMART MEDICAL DIAGNOSIS SYSTEM
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Initial Facts:
Fever = True
Cough = True
Rash = False
Breathlessness = True

--- Forward Chaining ---
Rule Fired: Fever AND Cough -> Flu
Rule Fired: Cough AND Breathlessness -> Pneumonia

Possible Diagnoses:
Flu = TRUE
Measles = Not Derivable
Pneumonia = TRUE

--- Backward Chaining ---
Goal: Pneumonia
Required conditions: Cough AND Breathlessness
All conditions are TRUE.
Therefore Pneumonia is PROVED.

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CASE STUDY 2 - AUTONOMOUS TRAFFIC MANAGEMENT AGENT
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Initial Facts:
Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA,Ambulance)

--- Unification ---
Rule 1:
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--- Unification ---
Rule 1:
Vehicle(x) AND EmergencyType(x) -> ClearPath(x)

Matching Vehicle(x) with Vehicle(Ambulance)
MGU = {'x': 'Ambulance'}

Derived:
ClearPath(Ambulance)

Rule 2 Fired:
GreenSignal(Ambulance)
NOT RedSignal(Ambulance)

Rule 3 Fired:
Proceed(CarA)

--- Resolution Proof ---
Goal: Proceed(CarA)
Negated Goal: NOT Proceed(CarA)

CNF Clauses:
C1: NOT Vehicle(x) OR NOT EmergencyType(x) OR ClearPath(x)
C2: NOT ClearPath(x) OR GreenSignal(x)
C3: NOT ClearPath(x) OR NOT RedSignal(x)
C4: NOT Vehicle(x) OR NOT Behind(x,y) OR NOT GreenSignal(y) OR Proceed(x)

Resolution steps:
Vehicle(Ambulance)
EmergencyType(Ambulance)
        ↓
ClearPath(Ambulance)
        ↓
GreenSignal(Ambulance)
        ↓
Vehicle(CarA) + Behind(CarA,Ambulance)
        ↓
Proceed(CarA)
        ↓
Proceed(CarA) AND NOT Proceed(CarA)
        ↓
Empty Clause

Conclusion:
Proceed(CarA) is PROVED by Resolution.
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--- Safety Analysis ---
Rule 4: NOT Wumpus(x,y) AND NOT Pit(x,y) -> Safe(x,y)

The required negative facts are not available.
Therefore the agent cannot prove that [1,1], [1,2], or [2,2] is completely safe.

--- Path Evaluation ---
Proposed path:
[1,1] -> [1,2] -> [2,2]

Analysis:
[1,1]:
Breeze is present; possible Pit nearby.
[1,2]:
Pit is a possible location.
[2,2]:
Gold is present, but safety is not proven.

Final Result:
The complete path is NOT guaranteed to be safe.

======================================================================
OVERALL RESULTS
======================================================================

Case Study 1:
Flu = TRUE
Pneumonia = TRUE
Measles = NOT DERIVABLE

Case Study 2:
ClearPath(Ambulance) = TRUE
GreenSignal(Ambulance) = TRUE
Proceed(CarA) = TRUE

Case Study 3:
IrrigationNeeded = TRUE
ApplyDripMethod = TRUE

Case Study 4:
Gold = [2,2]
Possible Wumpus = [1,1], [1,3], [2,2]
Possible Pit = [1,2], [2,1]
Complete path safety = NOT PROVED
```


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Case Study 4:
Gold = [2,2]
Possible Wumpus = [1,1], [1,3], [2,2]
Possible Pit = [1,2], [2,1]
Complete path safety = NOT PROVED
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**********************************************************************
                    PROGRAM COMPLETED
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